

TeleShift: Seasonal Reorganization of Global Extreme Precipitation Synchronization Networks

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Abstract

Extreme precipitation events frequently co-occur across geographically distant regions through atmospheric teleconnection pathways, yet how these global linkages reorganize between seasons remains poorly characterized. This study constructs directed extreme precipitation synchronization networks for boreal winter (DJF) and summer (JJA) using 46 years of ERA5 reanalysis data (1980–2026), applying Directional Event Synchronization to binary extreme event series defined by local 90th percentile thresholds. Edge significance is assessed against a surrogate null distribution, and three hypotheses regarding seasonal network structure are tested: whether winter connectivity exceeds summer, whether hub regions shift geographically between seasons, and whether long-range links are more abundant in winter.

All three hypotheses are supported at $p < 0.001$. DJF networks contain 93% more directed edges than JJA, hub regions shift 21.3° northward in winter driven by the activation of Northern Hemisphere midlatitude storm tracks, and 71% of winter links span more than 2,500 km compared to 56% in summer. Contrary to expectations, the Southern Ocean dominates hub activity in both seasons while the tropics exhibit near-zero hub presence throughout the year — a result inconsistent with the tropical monsoon dominance reported in prior literature. Furthermore, DJF is characterized by active cross-hemispheric coupling between Northern Hemisphere storm tracks and the Southern Ocean, whereas in JJA the dominant teleconnection flows are confined within the Southern Ocean — with only weak residual connections to the North Atlantic storm track. These findings reveal a fundamental seasonal dichotomy between a globally interconnected winter regime and a regionally fragmented summer regime in extreme precipitation synchronization.

Keywords: climate networks; event synchronization; extreme precipitation; teleconnections; seasonal variability; Southern Ocean; storm tracks; network science